Introduction

The following report's objective is to study the relation between the emission of nitrous oxide (NOX) and the measurement of nine sensors planted in a gas station. In this task, I will build a model to predict the emissions of nitrous oxide (NOX) given the data generated by the other nine sensors. This is considered a regression problem because we want to predict the values of a continuous dependent variable (NOX) based on the other nine independent values (sensors) hence using a regression algorithm is the appropriate method to solve this problem. Overall, I will consider building a Linear Regression model as the baseline model. I am going to use Mean Squared Error (MSE) as the performance metric for this problem because it measures the average squared errors difference between the predicted values and the actual values. Hence, evaluate the robustness to choose the best model in terms of generalising to unseen data.

Getting the data ready to be used with the machine learning model

Before building the models, I will proceed with a data preprocessing, to ensure the quality of our data, this include the following steps:

- First, I will remove the outliers, with the calculation of the upper and lower bound for each feature, to reduce the variance hence improve the model prediction accuracy.

```
# Remove outliers

for x in processed_data.columns:
    # Calculate the first and the last quartile
    q75,q25 = np.percentile(processed_data.loc[:,x],[75,25])
    # Calculate the upper and lower bounds
    intr_qr = q75-q25
    max = q75+(1.5*intr_qr) # upper bound
    min = q25-(1.5*intr_qr) # lower bound
    # Remove the rows that are higher than the upper bound or lower than lower bound
    processed_data.loc[processed_data[x] < min,x] = np.nan
    processed_data.loc[processed_data[x] > max,x] = np.nan
processed_data = processed_data.dropna(axis = 0)
```

Figure 1: Removing Outliers

- Second, I will standardize the independent variables (sensors) to remove the effect of the units using the MinMaxScaler() function:

```
# Standardize our data
scaler = preprocessing.MinMaxScaler()
X_scaled= scaler.fit_transform(X)
```

Figure 2: Standardizing Independent Variables

Methods

As shown in figure 3 below, we can notice from the correlation matrix that the correlation between every two variables is weak hence there is no linear relationship between the independent variables (sensors) and dependent variable (NOX) In spite of the low correlation between (NOX) and the other nine features as shown in the figure below, I will proceed with a multivariate regression model since we have a multivariate problem with nine explicative variables.

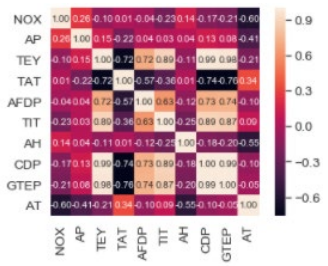


Figure 3: Correlation Matrix

To increase the robustness of the model, I chose to build a decision tree models that evaluate the model by averaging their performance across multiple runs or repeats of cross-validation. When fitting a final model, it may be desirable to either increase the number of trees or the max depth until the variance of the model is reduced across repeated evaluations, or to fit multiple final models and average their predictions. Scikit-learn library provides plenty of regression algorithms. In our case, I chose to use the three following regression algorithms:

- Linear Regression (Baseline Algorithm)
- Random Forest Regressor
- Gradient Tree Boosting Regressor

I chose Linear Regression model as a baseline model since it has a simple and decent performance for our problem. The point here it to try to improve the baseline model and measure its performance against Random Forest and Gradient Tree Boosting regressors models. I tried the following performance metrics:

1- **R Squared Error (R^2):** because it describes the proportion of variance explained by our model hence it can be used to know how well a regression model performs in general.
2- **Mean Squared Error (MSE):** because it tells us how close a regression line is to a set of points. It does this by taking the distances from the points to the regression line which represents the errors and squaring them. The squaring is necessary to remove any negative signs. The lower the value the better.

- **First Model: Linear Regression Model to Compare with Other Regressors**

```
np.random.seed(42) # to make the results reproducible

# Features
X = processed_data[['AT','AP','AH','AFDP','GTEP','TIT','TAT','TEY','CDP']]
# Target variable
Y = processed_data['NOX']

# Standardize our data
scaler = preprocessing.MinMaxScaler()
X_scaled= scaler.fit_transform(X)

xtrain, xtest, ytrain, ytest = train_test_split(X_scaled, Y, test_size=0.2)

reg = LinearRegression()
reg.fit(xtrain, ytrain)

# Make predictions using our regression model
ypred = reg.predict(xtest)

# Evaluate the regression model
print('--Baseline Linear Regression model metrics on the test set--\n')
print(f'Coefficient of Determination (R^2): {r2_score(ytest, ypred)*100:2f}%')
print(f'Mean Squared Error (MSE): {mean_squared_error(ytest, ypred)}')
```

```
--Baseline Linear Regression model metrics on the test set--

Coefficient of Determination (R^2): 77.278869%
Mean Squared Error (MSE): 18.111820675035005
```

Figure 4: Linear Regression Model (Baseline Model)

Before implementing the baseline model, I have splitted the dataset into training and testing sets to calculate the accuracy of our model. Next, I Instantiated the model and fit it to the training dataset. Finally, I made predictions on the testing dataset and calculated model performance. We notice a decent R^2 value (77.27%) and a bit high mean squared error (18.11) that indicates adequate performance of the baseline model.

- **Second Model: Random Forest Regressor model (Default Parameters)**

```
np.random.seed(42)
# Features
X = processed_data[['AT','AP','AH','AFDP','GTEP','TIT','TAT','TEY','CDP']]
# Target variable
Y = processed_data['NOX'].to_numpy()

# Standardize our data
scaler = preprocessing.MinMaxScaler()
X_scaled= scaler.fit_transform(X)

xtrain, xtest, ytrain, ytest = train_test_split(X_scaled, Y, test_size=0.2)


rf_base = RandomForestRegressor()
rf_base.fit(xtrain, ytrain)

# Make predictions using our regression model
ypred_rf_base = rf.predict(xtest)

# Evaluate the regression model
print('Random Forest Regression model metrics on the test set: \n')
print(f'Coefficient of Determination (R^2): {r2_score(ytest, ypred_rf_base)*100:2f}%')
print(f'Mean Squared Error (MSE): {mean_squared_error(ytest, ypred_rf_base)}')
```

```
Random Forest Regression model metrics on the test set:

Coefficient of Determination (R^2): 86.658999%
Mean Squared Error (MSE): 10.634585390876609
```

Figure 5: Random Forest Regressor Code Chunk

I implemented Random Forest Regressor model on the dataset with its default hyperparameters values.

I used the default hyperparameters values to check the model performance before improving it. The mean squared error of the predictions has decreased from (18.11) to (10.63) and R^2 has increased to (86.65%) which is a better performance than the baseline model.

- **Third Model: Gradient Boosting Regressor Model**

```
np.random.seed(42) # to make the results reproducible

# Features
X = processed_data[['AT','AP','AH','AFDP','GTEP','TIT','TAT','TEY','CDP']]
# Target variable
Y = processed_data['NOX']

# Standardize our data
scaler = preprocessing.MinMaxScaler()
X_scaled= scaler.fit_transform(X)

xtrain, xtest, ytrain, ytest = train_test_split(X_scaled, Y, test_size=0.2)


# Set GradientBoostingRegressor model and fit the model to training data
# ls: least squared error
gb_base = GradientBoostingRegressor()
gb_base.fit(xtrain, ytrain)

# Make predictions using our regression model
ypred_gb_base = gb_base.predict(xtest)

# Evaluate the regression model
print('Random Forest Regression model metrics on the test set: \n')
print(f'Coefficient of Determination (R^2): {r2_score(ytest, ypred_gb_base)*100:2f}%')
print(f'Mean Squared Error (MSE): {mean_squared_error(ytest, ypred_gb_base)}')
```

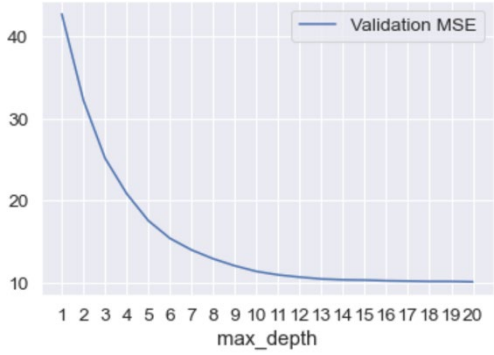
```
Random Forest Regression model metrics on the test set:

Coefficient of Determination (R^2): 84.224524%
Mean Squared Error (MSE): 12.575192564766414
```

Figure 6: Gradient Boosting Regressor Code Chunk

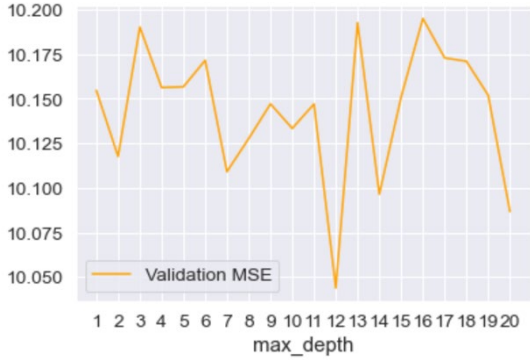
The performance of Gradient Boosting Regressor model on the dataset with its default hyperparameters values shows less performance than the Random Forest Model. Its mean squared error (12.57) and R^2 (84.22%).

<u>Tuning Hyperparameters</u>

In this case, I decided to improve models performance more by tuning the hyperparameter (max_depth) which represents the maximum depth of the tree and its value has an impact on the model performance. After splitting the training data into training and validation sets and doing Cross-validation to ensure that our model can generalize well. The result of evaluating the performance for both the two models based on MSE metric showed the optimal value of max_depth equals 20 (Random Forest Model) and max_depth of equals 12 (Gradient Boosting Model) as shown in below figures. These values decrease the MSE value hence improve the models performance in terms of generalising to unseen data.



Random Forest Regressor



Gradient Boosting Regressor

Results

Below are the testing results of the three models that show the performance of the models to generalise to unseen data.

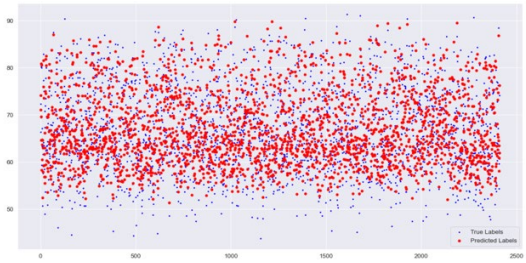


Figure 7: Baselline Model Results

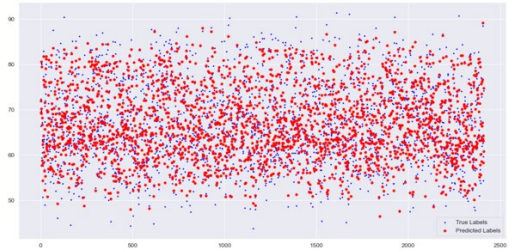


Figure 8: Tuned Random Forest Regressor Model Results (max_depth = 20)

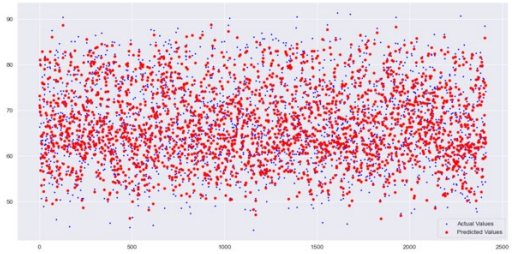


Figure 9: Tuned Gradient Boosting Regressor Model Results (max_depth=12)

From the figures above (Blue – Ground Truth, Red – Predicted Values), we notice that the baseline model mispredict a lot of nitrous oxide values in the bottom. Other tuned models appear to perform better and predict correctly some of these values in the bottom. Overall, tuned random forest regressor performs the best out of all of the three models.

- Baseline Model: MSE (18.11)
- Tuned Random Forest Model: MSE (10.60)
- Tuned Gradient Descent model: MSE (10.61)

Introduction

The objective of this study is to predict the demand for bicycles hiring (i.e., low, medium or high). The prediction is based on three categorical features in the dataset, i.e., "Season", "Workingday", and "Weathersit". The prediction model is developed using a Decision Tree.

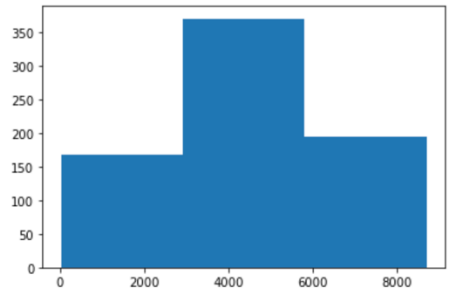
The structure of this report is as follows:

1. The dependent variable will be presented using a histogram to unscrew it into three groups (low, medium and high).
2. An entropy calculation function is developed and presented using the previously mentioned features.
3. A function to calculate the information gain (IG) is developed and presented.
4. A decision tree function is developed following the ID3 algorithm, and a sample decision tree is built and presented.
5. A new attribute called "tempbins" is calculated and used to build another tree.

Histogram of the counts

The following histogram represents the attribute "count" distribution, which reflects the demand for bicycles hire. As shown in the figure below, The values are divided into high, medium, and low based on this distribution ([22. , 2919.33333333, 5816.66666667, 8714.]). I have added a new feature in the dataset called 'usage', which contains values 'high', 'medium', and 'low' depending on these values distribution in the 'count' attribute.

```
count = df['count']
#TODO: fix the spaces
plt.hist(count, bins=3)  # 3 bins for three groups (high, medium, low)
plt.show()
```



Histogram of the count feature distribution

```
hist, bins = np.histogram(count, bins=3)
bins
# count
# hist
```

```
array([  22.        , 2919.33333333, 5816.66666667, 8714.        ])
```

Entropy and Information Gain Functions

First, I calculated the probabilities (probability for each category = number of occurrences per category/Variable size). Next, I calculated the entropy for each category and obtained the total entropy. The Entropy value equals (1.49). For the information gain (IG), the "Season" attribute has the highest information (0.25) among other attributes. The figure below shows the calculated entropy and IG.

```python
#entropy function
def entropy(d):
    # calculate probability for each type as number of occurrences / array length
    probabilities = [n_x / len(d) for x, n_x in collections.Counter(d).items()]

    # calculate per-character entropy fractions
    e_x = [-p_x * math.log(p_x, 2) for p_x in probabilities]

    # sum fractions to obtain Shannon entropy
    entropy = sum(e_x)
    return entropy
```

```python
# d = df['usage']
entropy(df.usage)
```

```
1.4917401149421705
```

Entropy function

```python
# Calculate information gain for each attribute
print('Information gain for Season: ', gain(df['season']))
print('Information gain for Weathersit: ', gain(df['weathersit']))
print('Information gain for Workingday: ', gain(df['workingday']))
```

```
Information gain for Season:  0.25248628103815296
Information gain for Weathersit:  0.0706214044037512
Information gain for Workingday:  0.0044543496239706215
```

Information Gain values for the three attributes ("Season", "Workingday", and "Weathersit")

Decision Tree Algorithm

```
{'season': {1: {'workingday': {0.0: {'weathersit': {1.0: 'low',
                                                    2.0: 'low',
                                                    3.0: 'low'}},
                               1.0: {'weathersit': {1.0: 'low',
                                                    2.0: 'low',
                                                    3.0: 'low'}}}},
            2: {'weathersit': {1.0: {'workingday': {0.0: 'medium',
                                                    1.0: 'medium'}},
                               2.0: {'workingday': {0.0: 'medium',
                                                    1.0: 'medium'}},
                               3.0: 'low'}},
            3: {'weathersit': {1.0: {'workingday': {0.0: 'medium',
                                                    1.0: 'medium'}},
                               2.0: {'workingday': {0.0: 'medium',
                                                    1.0: 'medium'}},
                               3.0: {'workingday': {0.0: 'medium',
                                                    1.0: 'low'}}}},
            4: {'weathersit': {1.0: {'workingday': {0.0: 'medium',
                                                    1.0: 'medium'}},
                               2.0: {'workingday': {0.0: 'medium',
                                                    1.0: 'medium'}},
                               3.0: {'workingday': {0.0: 'low', 1.0: 'low'}}}}}}
```

A Decision Tree built using the three attributes ("Season", "Workingday", and "Weathersit")

Overall, the decision tree structure has some redundant nodes. This can be improved by introducing a pruning technique to the tree. Pruning will eliminate unnecessary splits in the tree, keeping only the nodes that generalise better. Hence, reducing the tree's complexity and improving its accuracy (avoids miss-classification).

Adding a new attribute (tempbins) to the Decision Tree

This "tempbins" is calculated by binning the attribute temp into three values, i.e., high, medium and low. I've noticed that the attribute has a higher IG than other features in the dataset (0.2888); hence, it will alter the structure of the tree and generalises better than other features. Introducing a pruning technique can produce a better tree structure with higher classification accuracy.

```
{'tempbins': {'high': {'weathersit': {1.0: {'season': {2.0: {'workingday': {0.0: 'high',
                                                                            1.0: 'high'}},
                                                        3.0: {'workingday': {0.0: 'medium',
                                                                            1.0: 'medium'}},
                                                        4.0: {'workingday': {1.0: 'high'}}}},
                                          2.0: {'season': {2.0: {'workingday': {0.0: 'medium',
                                                                            1.0: 'medium'}},
                                                        3.0: {'workingday': {0.0: 'medium',
                                                                            1.0: 'medium'}},
                                                        4.0: {'workingday': {0.0: 'medium',
                                                                            1.0: 'medium'}}}},
                                          3.0: {'season': {3.0: {'workingday': {0.0: 'medium',
                                                                            1.0: 'low'}},
                                                        4.0: 'medium'}}}},
              'low': {'season': {1.0: {'weathersit': {1.0: {'workingday': {0.0: 'low',
                                                                            1.0: 'low'}},
                                                        2.0: {'workingday': {0.0: 'low',
                                                                            1.0: 'low'}},
                                                        3.0: 'low'}},
                                 2.0: 'low',
                                 4.0: {'workingday': {0.0: {'weathersit': {1.0: 'low',
                                                                            2.0: 'low',
                                                                            3.0: 'low'}},
                                                        1.0: 'medium'}}}},
              'medium': {'weathersit': {1.0: {'season': {1.0: {'workingday': {0.0: 'medium',
                                                                            1.0: 'medium'}},
                                                        2.0: {'workingday': {0.0: 'medium',
                                                                            1.0: 'medium'}},
                                                        3.0: {'workingday': {0.0: 'high',
                                                                            1.0: 'high'}},
                                                        4.0: {'workingday': {0.0: 'medium',
                                                                            1.0: 'medium'}}}},
                                          2.0: {'season': {1.0: {'workingday': {0.0: 'low',
                                                                            1.0: 'low'}},
                                                        2.0: {'workingday': {0.0: 'medium',
                                                                            1.0: 'medium'}},
                                                        3.0: {'workingday': {0.0: 'medium',
                                                                            1.0: 'medium'}},
                                                        4.0: {'workingday': {0.0: 'medium',
                                                                            1.0: 'medium'}}}},
                                          3.0: 'low'}}}}
```

Decision Tree Structure after adding the 'tempbins' attribute

References

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https://towardsdatascience.com/entropy-and-information-gain-in-decision-trees-c7db67a3a293

https://towardsdatascience.com/a-guide-to-decision-trees-for-machine-learning-and-data-science-fe2607241956

https://dhirajkumarblog.medium.com/decision-tree-from-scratch-in-python-629631ec3e3a

https://github.com/sirily11/MSBD5002-course-code/blob/1172c0b62b41f4e8b8b43448a31135e4668fc000/classification/decision_tree.py

https://www.kdnuggets.com/2020/01/decision-tree-algorithm-explained.html

I. Motivation

1. For what purpose was the dataset created?

The WikiArt Emotions dataset was created to study art and its impact on human's perception hence there is a need to improve the search for art with different features primarily the emotions they evoked on the observer.

2. Who created the dataset (e.g., which team, research group) and on behalf of which entity (e.g., company, institution, organisation)?

The dataset was created by Dr Saif M. Mohammad and Dr Svetlana Kiritchenko on behalf of the National Research Council (NRC) in Canada. Their research interests primarily in Computational Linguistics and Natural Language Processing (NLP).

3. Who funded the creation of the dataset?

The creation of the WikiArt Emotions dataset was conducted at the National Research Council (NRC) which is an organisation funded by the Canadian government.

II. Composition

1. What do the instances that compromise the dataset represent (e.g., documents, photos, people, countries)?

The WikiArt Emotions dataset consists of various pieces of art mostly paintings along with their crowdsourced annotations for emotions evoked in the observer. The actual pieces of art are selected from WikiArt.org's collection for four western styles which are Renaissance Art, Post-Renaissance Art, Modern Art, and Contemporary Art.

2. How many instances are there in total (of each type, if appropriate)?

There were 308 people that annotated between 20 and 1,525 pieces of art, where a total of 41,985 sets of responses were obtained.
There are 151,151 pieces of art in total available on the WikiArt.org website. These pieces of art are corresponding to ten main art styles and 168 style categories. The researchers have been able to annotate 4,105 pieces of art out of twenty-two categories on the WikiArt.org website.
Below is the list of the 22 categories with the number of art pieces annotated from each category.

- **Contemporary Art**

Minimalism	200
Modern Art	200
Abstract Art	200
Abstract Expressionism	200
Art Informel	20

Color Field Painting 200
Cubism 200
Expressionism 200
Impressionism 200
Lyrical Abstraction 200
Magic Realism 153
Neo-Expressionism 200
Pop Art 200
Post-Impressionism 200
Surrealism 198

- **Post-Renaissance Art**
  Baroque 200
  Neoclassicism 197
  Realism 200
  Rococo 200
  Romanticism 200

- **Renaissance Art**
  Early Renaissance 119
  High Renaissance 104
  Northern Renaissance 192

3. **Does the dataset contain all possible instances or is it a sample (not necessarily random) of instances from a larger set?**

   The WikiArt Emotions dataset is not a random sample. The style categories in the dataset were chosen for their popularity on the WikiArt.org website.

4. **What data does each instance consist of?**

   The instances consist of the responses to three questions asked for each piece of art. The annotations descriptions exactly as specified on the WikiArt Emotions Dataset website include:
   - All emotions that the image of art brings to mind
   - All emotions that the title of art brings to mind
   - All emotions that the art as a whole (title and image) brings to mind
   - A rating on a scale from -3 (dislike it a lot) to 3 (like it a lot)
   - Whether the image shows the face or the body of at least one person or animal
   - Whether the art is a painting or something else (e.g., sculpture).

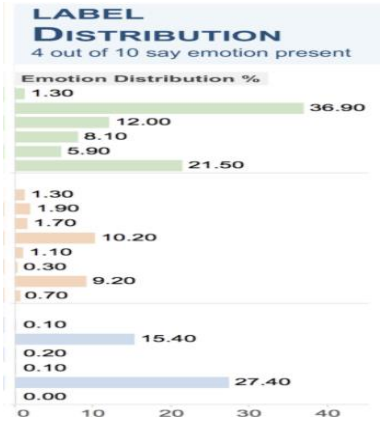
   Plus, the annotators have been asked a couple of questions to classify the emotions that the art elicits in three scenarios as specified on the website:
   Scenario I: we present only the image (no title), and ask the annotator to identify the emotions it evokes.

Scenario II: we present only the title of the art (no image), and ask the annotator to identify the emotions it evokes.
Scenario III: we present both the title and the image of the art and ask the annotator to identify the emotions that the art as a whole evokes.

5. **Is there a label or target associated with each instance?**

The researchers have created emotions annotations list consists of 20 emotions to identify what each piece of art evokes. Each piece of art was annotated with one or more of the twenty emotions as shown in the following figure:

Also, there is a representation that shows emotion distribution as shown in the following figure.

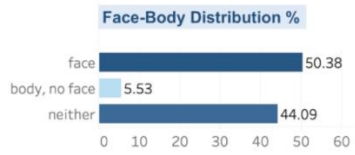


6. **Is the dataset self-contained, or does it link to or otherwise rely on external resources?**

   The WikiArt Emotions is not totally self-contained dataset. They do rely on external sources such as Image URL, Painting Info URL and Artist Info URL.

7. **Any other comments?**

   The art annotation doesn't only include emotion annotations but also includes the representation of a face and the average art rating, which indicates how much the observers enjoyed the art.



III. Collection Process

1. **How was the data associated with each instance acquired?**

   For each instance, or artwork, annotators were tasked with inputting their emotional response to the work under three scenarios, described in 2d. In total, 308 people annotated between 20 and 1,535 pieces of art.

2. **What mechanisms or procedures were used to collect the data?**

   The art was chosen based on what art was present on the WikiArt.org site. The categories that were explores were chosen based on how populous their categories were, and the paintings that were selected within these categories were chosen from WikiArt.org's 'Featured' tab.

   Annotations of the data were crowdsourced on the platform CrowdFlower. Annotations were verified by a gold-standard that had been curated by the authors based on several more objective questions (i.e., whether the artwork depicted a face or not) in the questionnaire.

3. **If the dataset is a sample from a larger set, what was the sampling strategy?**

   The sample of art to conduct the annotations from the WikiArt.org site was chosen based on how populous their corresponding styles were. The styles with more than 1000 paintings were chosen. For each chosen category, 200 paintings were selected from the 'Featured' tab for that category. Based on the researchers, these should "represent particularly significant pieces of art".

4. **Who was involved in the data collection processes and how were they compensated?** Crowdsourced participants on the CrowdFlower platform chose to complete annotations based on interest and compensation provided, though the compensation provided remains unclear.

5. **Were any ethical review processes conducted?**

   Due to hosting the annotation task on the CrowdFlower platform, all annotators had already agreed to the CrowdFlower terms of agreement.

   Additionally, "the annotation task was approved by the National Research Council Canada's Institutional Review Board under protocol 2017-98, which reviewed the proposed methods to ensure that they were ethical. Special attention was paid to obtaining informed consent and protecting participant anonymity."

IV. Preprocessing/cleaning/labelling

1. **Was any preprocessing/cleaning/labelling of the data done?**

   The WikiArt Emotions dataset was crowdsourced and cleaned using a gold instance checker. About 2% of the instances were internally annotated by the researchers and these gold instances were used to annotate other instances. The crowdsourced answer will be refused further annotation and discard all of their annotations if it fell below 70% when answering to a gold instance.

2. **Was the "raw" data saved in addition to the preprocessed/cleaned/labelled data?**

   According to the provided information by the researchers, they did not save any row data associated to the gold instance approach.

3. **Is the software used to preprocess/clean/label the instances available?**

   Yes, it is available publicly. They have done the annotations using software from a public crowdsourcing platform called CrowdFlower.

V. Uses

1. **Has the dataset been used for any tasks already?**

   The WikiArt Emotions dataset has been used in computer vision and natural language processing tasks specifically in information extraction to understand the content of images and text associated with them. This task enabled them to find that title often impacts the effectual response to art.

2. **Is there a repository that links to any or all papers or systems that use the dataset?**

   There is no public repository that links papers or systems that use the dataset.

3. **What (other) tasks could the dataset be used for?**

The WikiArt Emotions dataset can be used for various tasks such as automatic image processing and developing systems that detect emotions evoked by art. Regarding the lexicons, they can be used in commerce, public health, and research. For example, tracking brand and product perception via social media posts, tracking support for controversial issues and policies, tracking buy-in for non-pharmaceutical health measures such as social distancing during a pandemic, literary analysis, and developing more natural dialogue systems.

Anyone who wants to use the dataset must cite the paper below:

Saif M. Mohammad and Svetlana Kiritchenko. (2018). WikiArt Emotions: An Annotated Dataset of Emotions Evoked by Art. In Proceedings of the 11th edition of the Language Resources and Evaluation Conference, May 2018, Miyazaki, Japan.

4. **Is there anything about the composition of the dataset or the way it was collected and preprocessed/cleaned/labeled that might impact future uses?**

It might impact some downstream machine learning tasks that require a lot of data to learn from because the annotations were limited to only twenty emotional responses and the dataset size is small (4,105).

5. **Are there tasks for which the dataset should not be used?**

The dataset should not be used for commercial use without approval from Dr Saif M. Mohammad. Contact Saif M. Mohammad via saif.mohammad@nrc-cnrc.gc.ca and Pierre Charron via Pierre.Charron@nrc-cnrc.gc.ca
Also, the researchers must be cited properly when posting online or in news articles.

VI. Distribution

1. **Will the dataset be distributed to third parties outside of the entity on behalf of which the dataset was created?**

The WikiArt Emotions dataset is not allowed to be redistributed. Direct interested parties to this page: http://saifmohammad.com/WebPages/wikiartemotions.html

2. **How will the dataset be distributed?**

The WikiArt Emotions dataset can be obtained from DR Saif M. Mohammad's website: http://saifmohammad.com/WebPages/wikiartemotions.html.
The current distribution consists of the following data files:

"WikiArt-Emotions-Ag3.tsv", "WikiArt-Emotions-Ag4.tsv", "WikiArt-Emotions-Ag5.tsv", "WikiArt-Emotions-All.tsv"

Each file provides the infomation on the art taken from the WikiArt.org website:
- ID
- Style: Renaissance Art, Post-Renaissance Art, Modern Art, or Contemporary Art
- Category: one or more of the 22 style categories (if the art is assigned to more than one category, the categories are separated by a comma)
- Artist
- Title
- Year of creation
- Image URL
- Painting Info URL
- Artist Info URL

3. When will the dataset be distributed?

The WikiArt Emotions dataset has been distributed in August 2018 and still available on the website:
http://saifmohammad.com/WebPages/wikiartemotions.html.

4. Will the dataset be distributed under a copyright or other intellectual property (IP) license, and/or under applicable terms of use (ToU)?

The WikiArt Emotions Dataset has been made available in August 2018 and Copyrighted by National Research Council Canada (NRC).
It is available for research use except for commercial use, including for news articles, online posts, or product applications, users are asked to contact Said M. Mohammad.

Terms of use:
a) If you use this dataset, cite the paper below:

Saif M. Mohammad and Svetlana Kiritchenko. (2018). WikiArt Emotions: An Annotated Dataset of Emotions Evoked by Art. In Proceedings of the 11th edition of the Language Resources and Evaluation Conference, May 2018, Miyazaki, Japan.

b) Do not redistribute the data. Direct interested parties to this page:
http://saifmohammad.com/WebPages/wikiartemotions.html

c) National Research Council Canada (NRC) disclaims any responsibility for the use of the dataset and does not provide technical support. However, the contact listed above will be happy to respond to queries and clarifications.

5. **Have any third parties imposed IP-based or other restrictions on the data associated with the instances?**

   These are the restrictive uses as described on the WikiArt Website:

   - If interested in commercial use of any of these lexicons, send email to Saif M. Mohammad (Senior Research Officer at NRC and creator of these lexicons): saif.mohammad@nrc-cnrc.gc.ca and Pierre Charron (Client Relationship Leader at NRC): Pierre.Charron@nrc-cnrc.gc.ca. A nominal one-time licensing fee may apply.

   - In news articles and online posts on work using these lexicons, cite the appropriate lexicons. For example: "This application/product/tool makes use of the <resource name>, created by <author(s)> at the National Research Council Canada." (The creators of each lexicon are listed below. Also, if you send us an email, we will be thrilled to know about how you have used the lexicon.) If possible hyperlink to this page: http://saifmohammad.com/WebPages/lexicons.html

   - If you use a lexicon in a product or application, then acknowledge this in the 'About' page and other relevant documentation of the application by stating the name of the resource, the authors, and NRC. For example: "This application/product/tool makes use of the <resource name>, created by <author(s)> at the National Research Council Canada." (The creators of each lexicon are listed below. Also, if you send us an email, we will be thrilled to know about how you have used the lexicon.) If possible hyperlink to this page: http://saifmohammad.com/WebPages/lexicons.html

   - Do not redistribute the data. Direct interested parties to this page: http://saifmohammad.com/WebPages/lexicons.html

   - National Research Council Canada (NRC) disclaims any responsibility for the use of the lexicons listed here and does not provide technical support. However, the contact listed above will be happy to respond to queries and clarifications.

VII. Maintenance

1. **Who is supporting/hosting/maintaining the dataset?**

   The WikiArt Emotions dataset is maintained by the researcher Dr Saif Mohammad on his website. The dataset can be found on this link: http://saifmohamad.com/WebPages/wikiartemotions.html

2. **How can the owner/curator/manager of the dataset be contacted?**

   Anyone who has any query about the WikiArt Emotions dataset can contact one of the following:

   - Dr Saif Mohammad
     Researcher in Computational Linguistics at the National Research Council Canada (NRC)
     Email: saif.mohammad@nrc-cnrc.gc.ca
     phone: +1-613-993-0620
   - Dr Svetlana Kiritchenko
     Senior Research Scientist, National Research Council Canada
     Email: svetlana.kiritchenko@nrc-cnrc.gc.ca

3. **Is there an erratum?**

   The dataset does not appear to have any erratum.

4. **Will the dataset be updated?**

   They are no details mentioned whether the dataset will be updated in the future. Nonetheless, it appears that if there are any update on the dataset, it should be reflected here: http://saifmohammad.com/WebPages/wikiartemotions.html

5. **Will older versions of the dataset continue to be supported/maintained?**

   The researchers did not mention whether they will continue supporting older versions of the WikiArt Emotions dataset.

6. **If others want to extend/augment/build on/contribute to the dataset, is there a mechanism for them to do so?**

   The researchers will be able to help in answering any enquiries about the project. Still, they do not redistribute the art images, they provide only the annotations and disclaim any responsibility for the use of the lexicons and does not provide any technical support.